A Unified Physical Consumer-Resource Neural ODE Framework for Microbial Serial Dilution Ecology Across Diverse Microbiomes

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October 22, 2026

Abstract

Microbial community dynamics in serial dilution regimes are governed by continuous within-passage substrate uptake, metabolic cross-feeding, and density-dependent crowding, interrupted by periodic discrete dilution and dispersal events. Traditional modeling paradigms rely either on high-parameter phenomenological Generalized Lotka-Volterra matrices that suffer from severe overfitting, or on discrete multi-step Markov approximations that obscure intra-passage kinetics and distort nonlinear regulation via artificial simplex resets. Here, we present a unified **Physical Consumer-Resource Neural Ordinary Differential Equation (Neural ODE)** framework that integrates low-dimensional latent resource niches ($K \ll N$), thermodynamic non-negativity constraints, quadratic self-limitation, and unnormalized physical biomass continuity across dilution passages. We demonstrate the universality of this model across four benchmark empirical serial dilution datasets spanning soil, mammalian rumen, and human gut microbiota: (i) 48 soil and gut isolates under 3 nutrient regimes with daily $1:30$ dilution and 10^{-6} regional dispersal (Gore et al., *Science* 2022); (ii) 50 human gut isolates across 5 spatial carrier microenvironments with $1:100$ dilution (Pollard / Jin et al., *Nat. Commun.* 2023); (iii) 32 rumen microbiome taxa under 6 high-pressure and chemical forcing regimes across 2 donor hosts (German et al.); and (iv) 12 synthetic gut species across 175 combinatorial designs under exact $1:20$ serial transfers with dual-phase intra-passage sampling (Venturelli et al., *Mol. Syst. Biol.* 2018). Our physical ODE framework achieves superior predictive accuracy across all datasets ($R^2 = 0.67$–0.97), resolves the simplex reset artifact by boosting empirical R^2 by up to $+9.4\%$, and sustains stable multi-species coexistence across 100 simulated transfers without artificial extinction collapses.

1 Introduction

Microbial communities in nature and industrial bioprocesses rarely exist in isolated steady-state chemostats. Instead, they are subjected to periodic boom-and-bust cycles driven by batch growth, substrate depletion, and periodic dilution or dispersal [1, 2]. Laboratory serial dilution experiments have become the quintessential model system to study these non-equilibrium ecological dynamics under controlled conditions [3–5]. In a typical serial dilution regime, a defined or complex microbial inoculum is allowed to grow in a batch culture for an incubation period τ (typically 24 to 48 hours), after which a fixed fraction $1/D$ is transferred into fresh nutrient medium, repeating across multiple passages.

Despite the conceptual simplicity of serial transfers, modeling and predicting the multi-passage trajectories of multispecies consortia remains a formidable challenge:

1. **High Dimensionality and Overfitting:** Classical Generalized Lotka-Volterra (gLV) models describe inter-species interactions via an interaction matrix $A \in \mathbb{R}^{N \times N}$. For realistic communities of $N = 32$ to 50 species, this entails learning 1000 to 2500 parameters, inevitably leading to severe overfitting and catastrophic error propagation over long multi-passage rollouts.

2. **Phenomenological vs. Physical Mechanisms:** gLV models treat interactions as direct species-species pairwise forces, failing to capture substrate exhaustion, metabolic niche partitioning, and resource-mediated cross-feeding [6–8].

3. **Discontinuous vs. Continuous Coupling:** A serial dilution experiment consists of continuous physical growth and resource uptake within each passage, interrupted by discrete dilution and dispersal events at passage boundaries. Naive machine learning models often discretize passages as Markov steps,

completely obscuring intra-passage growth kinetics, saturation limits, and transient phases.

4. **Simplex Normalization Artifacts:** Previous modeling efforts often normalized biomass vectors to the unit simplex at every transfer ($\sum_i x_i = 1$). While convenient for relative sequencing read counts, this unphysical operation severely alters the mathematical magnitude of nonlinear self-limitation terms ($-\alpha_i N_i^2$), artificially compressing biomass by several orders of magnitude and muting negative feedback mechanisms necessary for stable coexistence.

To overcome these foundational limitations, we developed a unified **Physical Consumer-Resource Neural Ordinary Differential Equation (Neural ODE)** framework [9] specifically tailored to serial dilution ecology. By embedding low-dimensional latent resource niches ($K \ll N$) with non-negative metabolic fluxes, quadratic self-limitation, and authentic physical biomass propagation across dilution transfers, our framework accurately captures community assembly across four distinct empirical datasets:

- **Gore et al. (Science 2022)**: 48 soil and gut isolates across 99 synthetic communities under 3 nutrient regimes with daily $1:30$ dilution and 10^{-6} regional dispersal [3].
- **Pollard / Jin et al. (Nature Communications 2023)**: 50 human gut isolates cultivated across 5 spatial carrier microenvironments with $1:100$ daily dilution [4].
- **German et al. (Rumen Enrichment)**: 32 core rumen microbiome taxa tracked under 6 high-pressure and methanogenic substrate regimes over 6 transfer cycles.
- **Ophelia / Venturelli et al. (Molecular Systems Biology 2018)**: 12 synthetic gut species across 175 combinatorial designs subjected to $1:20$ serial transfers with dual-phase (mid-log and stationary) intra-passage sampling [5].

In this paper, we present the unified mathematical model, describe how each experimental dataset was rigorously parameterized and modeled, report cross-dataset predictive accuracy gains over legacy baselines, and demonstrate long-term ecological stability and diversity retention.

2 Mathematical Formulation: Physical Consumer-Resource Neural ODE

We model microbial community dynamics in serial dilution regimes through a coupled hybrid dynamical system combining continuous-time physical ordinary differential equations within each incubation cycle with discrete reset maps at transfer boundaries.

2.1 Continuous-Time Intra-Passage Dynamics ($0 \le t \le \tau$)

Let $N_i(t) \ge 0$ denote the physical biomass density of microbial taxon $i \in \{1, \dots, N\}$, and let $R_k(t) \ge 0$ denote the concentration of latent environmental resource niche $k \in \{1, \dots, K\}$, where $K \ll N$. Within each incubation period of duration τ, the system evolves according to the coupled autonomous ODEs:

$$\frac{dN_i}{dt} = N_i(t) \left(\sum_{k=1}^{K} \Theta_{ik} R_k(t) \right) - \alpha_i N_i(t)^2, \tag{1}$$

$$\frac{dR_k}{dt} = -R_k(t) \sum_{l=1}^{K} W_{kl} \left(\sum_{j=1}^{N} \Theta_{jl} N_j(t) \right). \tag{2}$$

2.1.1 Biological and Physical Interpretation of Parameters

- **Consumer Uptake Matrix $\Theta \in \mathbb{R}_{\ge 0}^{N \times K}$:** Θ_{ik} represents the metabolic uptake affinity and growth yield of species i on resource niche k. To enforce strict thermodynamic and biological validity, non-negativity is guaranteed via a smooth transformation:

$$\Theta = \text{softplus}(\Theta_{\text{raw}}) = \log(1 + \exp(\Theta_{\text{raw}})). \tag{3}$$

- **Metabolic Interaction Matrix $W \in \mathbb{R}_{\ge 0}^{K \times K}$:** W_{kl} models resource transformation, cross-feeding, and secondary metabolite coupling between latent niches. Non-negativity $W = \text{softplus}(W_{\text{raw}})$ ensures that resource consumption fluxes remain strictly dissipative and mass-conserving.
- **Quadratic Self-Limitation $\alpha \in \mathbb{R}_{>0}^{N}$:** The term $-\alpha_i N_i^2$ represents intra-species crowding, physical space saturation, autotoxicity, and metabolic waste accumulation [6, 7]. Unlike phenomenological clipping or ad-hoc capacity ceilings, quadratic self-limitation naturally regularizes unbounded exponential explosion, producing smooth, biologically realistic logistic-type saturation as resources are drawn down. It is parameterized as $\alpha_i = \text{softplus}(\alpha_{\text{raw},i})$.

2.2 Discrete Transfer Map Between Passages ($m \to m+1$)

At the end of each passage $m \in \{1, \dots, M-1\}$, the culture reaches time $t = \tau$. An aliquot of fraction $1/D$ is sampled and diluted into fresh nutrient media:

$$N_i(0)^{(m+1)} = \left(\frac{N_i(\tau)^{(m)}}{D} + \mu \cdot M_{c,i} \right) \cdot M_{c,i}, \tag{4}$$

$$R_k(0)^{(m+1)} = R_{0,c,k}, \tag{5}$$

where:

- $D > 1$ is the physical dilution factor (e.g., $D = 30.0$ for Gore, $D = 100.0$ for Pollard and German, $D = 20.0$ for Venturelli).
- $\mu \geq 0$ is the regional dispersal / migration floor. In experiments where migration was artificially introduced by spiking a background regional stock (such as Gore et al., where $\mu = 10^{-6}$), $\mu > 0$ prevents non-physical absorbing-state extinctions. In closed systems (Pollard, German, Venturelli), $\mu = 0.0$ strictly.
- $M_{c,i} \in \{0, 1\}$ is the binary community composition mask indicating whether species i was inoculated into community c. Uninoculated species ($M_{c,i} = 0$) are strictly held at zero.
- $R_{0,c} = \text{softplus}(a_c) \in \mathbb{R}^K_{>0}$ is the unconstrained replenished environmental resource vector for experimental condition c, optimized via gradient descent.

2.3 Physical Biomass Continuity vs. The Simplex Reset Bug

A critical mathematical insight in this work concerns the propagation of biomass across transfers. In earlier microbial modeling literature, investigators frequently rescaled biomass to the unit simplex at every dilution passage:

$$N_i(0)^{(m+1)}_{\text{flawed}} = \frac{N_i(\tau)^{(m)}}{\sum_{j=1}^{N} N_j(\tau)^{(m)}} \cdot \frac{1}{D}. \tag{6}$$

While seemingly harmless for relative sequencing reads, equation (6) introduces a severe mathematical distortion:

- In a 50-species community, relative fractions satisfy $x_i \approx 0.02$.
- Consequently, the quadratic self-limitation term becomes:

$$-\alpha_i N_i^2 \approx -\alpha_i (0.02)^2 = -0.0004\alpha_i. \tag{7}$$

- At this suppressed scale, quadratic negative feedback is reduced by three to four orders of magnitude. The dynamics collapse into linear exponential selection where the single fastest-growing species excludes all competitors within 5 to 10 transfers (the classical Competitive Exclusion Principle).

By replacing equation (6) with true **unnormalized physical biomass continuity** (equation (4)), biomass maintains its natural magnitude ($N_i \sim 0.1$–2.0), restoring the physiological efficacy of $-\alpha_i N_i^2$ and enabling persistent multi-species coexistence across dozens of transfers.

2.4 Objective Functions and Loss Formulations

Depending on the experimental measurement modality of each dataset, we employ tailored, biophysically sound loss functions:

2.4.1 Unified Generalized KL Loss for Absolute Biomass

For datasets where total optical density OD_{600} is measured (Gore et al. and Venturelli et al.), absolute biomass is known: $y_{c,i}(m) = x_{c,i}(m) \cdot \text{OD}_c(m)$. To train without imposing homoskedastic Gaussian noise assumptions on positive-definite biomass, we employ Dr. Dixit's Generalized KL divergence:

$$\mathcal{L}_{\text{GKL}}(y, \hat{y}) = \sum_{c,m,i} \Big(\hat{y}_{c,i}(m) - y_{c,i}(m) + y_{c,i}(m) \log \left(\frac{y_{c,i}(m) + \epsilon}{\hat{y}_{c,i}(m) + \epsilon} \right) \Big), \tag{8}$$

where $\epsilon = 10^{-6}$ prevents numerical singularities. Mathematically, Generalized KL is the Bregman divergence corresponding to a Poisson dispersion process ($\text{Var}(y) \propto \hat{y}$), naturally down-weighting high-OD stationary-phase extremes and preserving accurate proportional growth slopes across lag and exponential phases.

2.4.2 Simplex Relative KL Divergence (Amplicon / Metagenomic Reads)

For datasets providing high-throughput sequencing reads without total biomass quantification (e.g., Pollard, German), the predicted biomass is projected onto the simplex:

$$\hat{x}_{c,i}(m) = \frac{\hat{N}_{c,i}(m)}{\sum_{j=1}^{N} \hat{N}_{c,j}(m) + \epsilon}. \tag{9}$$

The model is then trained against empirical relative fractions $x_{c,i}(m)$ using discrete relative entropy:

$$\mathcal{L}_{\text{KL}}(x, \hat{x}) = \sum_{c,m,i} x_{c,i}(m) \log \left(\frac{x_{c,i}(m) + \epsilon}{\hat{x}_{c,i}(m) + \epsilon} \right). \tag{10}$$

In German et al., equation (10) is evaluated strictly over the donor-accessible sub-simplex enforced by host membership masks $M_{c,i}$.

2.5 Analytical Derivation of the Environment-Dependent Effective Interaction Matrix

A central theoretical question in microbial ecology is whether continuous consumer-resource dynamics can be rigorously mapped to macroscopic inter-species interaction matrices of the Lotka-Volterra type, and whether such interactions remain invariant across different environmental regimes [6, 10]. In our framework, the per-capita growth rate of species i during continuous intra-passage cultivation is given by:

$$g_i(\mathbf{N}, \mathbf{R}) = \frac{1}{N_i} \frac{dN_i}{dt} = \sum_{k=1}^{K} \Theta_{ik} R_k(t) - \alpha_i N_i(t). \tag{11}$$

The effective inter-species interaction coefficient $A_{ij} \equiv \frac{\partial g_i}{\partial N_j}$, representing the marginal influence of species j's abundance on species i's per-capita fitness, is obtained by differentiating g_i with respect to biomass N_j:

$$A_{ij} = \sum_{k=1}^{K} \Theta_{ik} \frac{\partial R_k}{\partial N_j} - \alpha_i \delta_{ij}. \tag{12}$$

2.5.1 Resource Depletion and Cross-Feeding Mediated Coupling

The sensitivity $\frac{\partial R_k}{\partial N_j}$ quantifies how the growth of species j alters the availability of resource k. Integrating the continuous resource dynamics (2) over an incubation passage shows that the instantaneous resource concentration scales directly with the initial environmental nutrient endowment $\mathbf{R}_{0,c} = \text{softplus}(\mathbf{a}_c)$:

$$R_k(t) = R_{0,c,k} \exp\left(-\sum_{j=1}^{N} \left(\Theta_{jk} - \sum_{l=1}^{K} \Theta_{jl} W_{lk} \right) \int_0^t N_j(s) ds \right) \tag{13}$$

Differentiating the time-averaged resource availability over the passage with respect to species j yields:

$$\frac{\partial R_k}{\partial N_j} \approx -\kappa\, R_{0,c,k} \left(\Theta_{jk} - \sum_{l=1}^{K} \Theta_{jl} W_{lk} \right), \tag{14}$$

where $\kappa > 0$ represents the characteristic integration timescale. Substituting equation (14) into equation (12) yields the closed-form effective interaction matrix for experimental condition c:

$$A_{ij}^{(c)} = -\alpha_i \delta_{ij} - \kappa \sum_{k=1}^{K} \Theta_{ik} R_{0,c,k} \left(\Theta_{jk} - \sum_{l=1}^{K} \Theta_{jl} W_{lk} \right). \tag{15}$$

In compact matrix notation, the effective interaction operator $\mathbf{A}^{(c)} \in \mathbb{R}^{N \times N}$ decomposes into three distinct physical mechanisms:

$$\mathbf{A}^{(c)} = \underbrace{-\text{diag}(\alpha)}_{\text{Intra-specific crowding}} \underbrace{-\kappa\, \boldsymbol{\Theta}\, \text{diag}(\mathbf{R}_{0,c})\, \boldsymbol{\Theta}^T}_{\text{Niche overlap competition}} \underbrace{+\kappa\, \boldsymbol{\Theta}\, \text{diag}(\mathbf{R}_{0,c})\, \mathbf{W}^T \boldsymbol{\Theta}^T}_{\text{Metabolic cross-feeding facilitation}} \tag{16}$$

2.5.2 Key Properties and Environmental Modulation

Equation (16) reveals three fundamental properties that distinguish physical consumer-resource models from phenomenological Generalized Lotka-Volterra matrices:

1. **Environmental Nutrient Scaling:** The effective interaction matrix $\mathbf{A}^{(c)}$ is explicitly modulated by the external nutrient vector $\mathbf{R}_{0,c}$. Under low nutrient supplies ($\mathbf{R}_{0,c} \approx 0$), inter-species couplings vanish, and dynamics are dominated by intra-specific crowding (α_i). Conversely, nutrient enrichment linearly amplifies both competitive overlap and cross-feeding facilitation. In Gore et al., calculating $\mathbf{A}^{(c)}$ from the learned parameters reveals that the average off-diagonal interaction strength nearly doubles from 1.57 in Low Nutrients to 3.04 in High Nutrients, providing the direct physical driver for the emergent chaotic bifurcation at $S = 12$.

2. **Context-Dependent Sign Inversion (Competition vs. Facilitation):** In standard gLV models, the sign of A_{ij} is fixed. In equation (16), the net coupling between species i and j on resource k is proportional to $R_{0,c,k}(\sum_l \Theta_{jl} W_{lk} - \Theta_{jk})$. If the environment is depleted in metabolite k while primary substrate l is abundant, species j secretes k for species i, producing cooperative facilitation ($A_{ij} > 0$). If the medium is artificially supplemented with nutrient k, both species consume it directly, flipping the interaction to exploitative competition ($A_{ij} < 0$).

3. **Low-Rank Stability (rank $\leq K \ll N$):** Unlike unconstrained $N \times N$ interaction matrices that contain N^2 independent parameters, the off-diagonal interaction network is strictly constrained by the K latent resource dimensions:

$$\text{rank}\left(\mathbf{A}^{(c)} + \text{diag}(\alpha) \right) \leq K \ll N. \tag{17}$$

This low-rank metabolic constraint prevents destructive destabilizing feedback loops, explaining why high-diversity communities ($N = 48$ in Gore, $N = 91$ in German) evade the classic May-Wigner complexity-stability limit and maintain robust coexistence across 100 serial transfers.

3 Empirical Datasets and Experimental Modeling Protocols

To assess the universality and predictive accuracy of our Physical Consumer-Resource Neural ODE framework, we applied it to four distinct, high-impact microbial serial dilution datasets spanning soil, mammalian rumen, and human gut microbiota. Table 1 summarizes the physical parameters, experimental designs, and modeling configurations across all four studies.

3.1 Gore et al. (*Science* 2022): Emergent Phases of Ecological Dynamics

3.1.1 Biological Microcosms and Combinatorial Assembly

Hu et al. [3] constructed synthetic microcosms from a defined pool of 48 bacterial strains: Species 1–24 are soil isolates, and Species 25–48 are *Caenorhabditis elegans* gut isolates. Microcosms were assembled combinatorially at five levels of initial richness:

Table 1: Cross-Dataset Protocol and Modeling Parameter Comparison.

Dimension	**Gore et al. (2022)**	**Pollard / Jin et al. (2023)**	**German et al. (Rumen)**	**Venturelli et al. (2018)**
Biological System	Soil & gut isolates	Human gut isolates	Cow rumen fluid	Human gut core strains
Species Count (N)	48	50	91	12
Latent Niches (K)	2	5	12	2
Dilution Factor (D)	1 : 30 (30.0)	1 : 100 (100.0)	1 : 100 (100.0)	1 : 20 (20.0)
Transfer Cadence (τ)	24 h	24 h	Variable batch cycle	24 h
Sampling Horizon	Days 0–10 (10 transfers)	Passages 1–6 (6 transfers)	Transfers 1–6 (6 transfers)	$t \in \{0, 12, 24, 36, 48, 60\}$h
Data Modality	Absolute OD_{600} biomass	Relative read fractions	Relative 16S read fractions	Absolute OD_{600} biomass
Dispersal Floor (μ)	10^{-6} (Authentic lab pool)	0.0 (Strictly closed)	0.0 (Strictly closed)	0.0 (Strictly closed)
Nutrient / Media Regimes	3 (LN, MN, HN)	5 (Spatial carriers/broth)	6 (Pressure × Substrate)	1 (Anaerobic Basal Broth)
Environmental Vector R_0	Monotonic scalar ladder	Learned $\mathbb{R}^5$ per carrier	Learned $\mathbb{R}^4$ per condition	Learned $\mathbb{R}^2$ baseline
Loss Function	Generalized KL (GKL)	Simplex Relative KL	Simplex Relative KL	Hybrid MSE + 0.05 · KL
Community Inocula	99 ($S = 3, 6, 12, 24, 48$)	5 conditions (all 50 sp.)	12 (2 hosts × 6 cond.)	175 (Pairs, trios, dropouts)

- $S = 3$ (8 distinct communities, soil isolates only),
- $S = 6$ (8 distinct communities, soil isolates only),
- $S = 12$ (8 distinct communities, soil isolates only; the critical bifurcation point),
- $S = 24$ (8 distinct communities, mixed soil and gut isolates),
- $S = 48$ (1 community, full 48-species library).

Each of the 33 communities was cultivated in triplicate across three nutrient environments:

1. **Low Nutrient (LN)**: Base medium without supplementary carbon/nitrogen sources.
2. **Medium Nutrient (MN)**: Base medium supplemented with 5 g/L glucose and 4 g/L urea.
3. **High Nutrient (HN)**: Base medium supplemented with 20 g/L glucose and 16 g/L urea (4× scalar increase over MN).

3.1.2 Physical Transfer and Dispersal Modeling

Communities underwent daily 1 : 30 serial dilutions ($D = 30.0$) over 10 consecutive days. Crucially, the experimental protocol incorporated a continuous regional dispersal floor: each day, a background stock solution of all 48 species was introduced at a relative frequency of $\mu = 10^{-6}$. In our model, this is faithfully represented as:

$$N_i(0)^{(m+1)} = \left(\frac{N_i(\tau)^{(m)}}{30.0} + 10^{-6} \cdot M_{c,i} \right) \cdot M_{c,i}. \quad (18)$$

The environmental supply vectors were parameterized monotonically to mirror the chemical recipe:

$$\begin{aligned} R_0(\text{LN}) &= \text{softplus}(a_0), \\ R_0(\text{MN}) &= R_0(\text{LN}) + \text{softplus}(\Delta a_1), \\ R_0(\text{HN}) &= R_0(\text{MN}) + 4 \cdot \text{softplus}(\Delta a_2). \end{aligned} \quad (19)$$

Total biomass is tracked via OD_{600} and trained using the Generalized KL loss (equation (8)).

3.2 Pollard / Jin et al. (*Nat. Commun.* 2023): Spatial Structuring in Gut Consortia

3.2.1 Microbiome Consortia and Spatial Particulate Niches

Jin et al. [4] cultivated a complex synthetic consortium of 50 human gut bacterial isolates across 6 daily serial passages (1 : 100 dilution, $D = 100.0$) to investigate how physical spatial structures preserve taxonomic diversity. The consortia were grown in 5 physical carrier conditions:

1. Mucin-agar hydrogel beads,
2. Mucin bead supernatant (planktonic fraction),
3. Plain-agar hydrogel beads,
4. Plain bead supernatant,
5. Liquid broth control (no physical carriers).

3.2.2 Closed System Modeling and Simplex Reset Elimination

Because the Pollard experiment was completely closed without regional migration, the dispersal parameter is strictly set to zero ($\mu = 0.0$):

$$N_i(0)^{(m+1)} = \frac{N_i(\tau)^{(m)}}{100.0}. \tag{20}$$

The 5 environmental niches are represented by freely learned 5-dimensional resource supply vectors $R_{0,c} = \text{softplus}(a_c) \in \mathbb{R}^5$ ($c \in \{1, \ldots, 5\}$). As detailed in Section 2, previous models rescaled the biomass vector to the unit simplex at every transfer, which suppressed $-\alpha_i N_i^2$ by a factor of $(1/50)^2 = 0.0004$ and triggered severe extinction cascades. Propagating unnormalized physical biomass resolved this defect, accurately reproducing the high diversity retained in mucin carriers (21–24 surviving species) relative to liquid broth (11 species).

3.3 German et al.: Rumen Enrichment Under High Pressure and Chemical Forcing

3.3.1 Enrichment Regimes and Donor Inocula

The German et al. dataset tracks the selective enrichment of core anaerobic rumen microbial communities subjected to 6 distinct metabolic and physical forcing regimes:

- Headspace gas pressure: 5.0 bar vs. 30.0 bar (H_2 : CO_2),
- Methanogenic substrate supplementation: Methanol (MeOH) vs. Trimethylamine (TMA) vs. unsupplemented control.

These 6 conditions were inoculated in parallel using rumen fluid from two distinct donor cows (OSU and BRRS), generating 12 independent longitudinal trajectories tracked across 6 serial transfer cycles (T1 to T6). Rather than using an overly restrictive baseline filter that discarded 45 out of 46 methanogens, we apply a longitudinal peak abundance threshold of $\geq 1\%$ ($\max_{c,t} x_{c,t,i} \geq 0.01$). This retains $N = 91$ core taxa, comprising 22 methanogenic archaea (capturing 98.5% of all archaeal reads) and 69 bacteria spanning major fermenter guilds, achieving 92.3% total community coverage across all 72 samples.

3.3.2 Parameterization and Closed Dynamics

Serial transfers followed a 100-fold dilution factor ($D = 100.0$) with zero migration ($\mu = 0.0$). A 12-dimensional latent resource space ($K = 12$), selected by minimal validation loss, captures primary hydrogenotrophic, methylotrophic, and syntrophic fermentation pathways. Resource supplies $R_{0,c} = \text{softplus}(a_c) \in \mathbb{R}^{12}$ were optimized independently for each treatment. The model was trained using the relative simplex KL loss (equation (10)).

3.4 Ophelia / Venturelli et al. (*Mol. Syst. Biol.* 2018): High-Throughput Synthetic Gut Consortia

3.4.1 Combinatorial Assembly in Anaerobic Basal Broth

Venturelli et al. [5] characterized microbial interactions in a 12-species synthetic human gut consortium by systematically testing 175 combinatorial sub-communities in Anaerobic Basal Broth (ABB):

- 109 pairwise communities ($S = 2$),
- 33 trio communities ($S = 3$),
- 18 higher-order mixtures ($S \in \{4, 5, 6\}$),
- 15 dropout communities ($S \in \{11, 12\}$).

3.4.2 Authentic Serial Transfer Protocol and Multi-Phase Temporal Sampling

While earlier analyses sometimes treated the Venturelli dataset as an unperturbed continuous batch culture, audit of the primary experimental methods revealed that the experiment was conducted as a **24-hour serial transfer experiment with 20-fold dilution** ($D = 20.0$):

- Cultures were transferred into fresh ABB media at 24 h and 48 h using a 1 : 20 dilution factor.
- Optical density OD_{600} and individual species abundances were measured at **both** 12 **h (mid-exponential phase) and** 24 **h (stationary phase)** of each passage:

$$t_{\text{sample}} \in \{0 \text{ h}, 12 \text{ h}, 24 \text{ h}, 36 \text{ h}, 48 \text{ h}, 60 \text{ h}\}. \tag{21}$$

We parameterized this exact multi-phase sampling schedule inside the ODE solver. The initial biomass of each community was derived from the measured Day 0 OD_{600}, with zero migration ($\mu = 0.0$) and composition strictly masked by $M_{c,i}$. The model was trained via the hybrid objective (equation (**??**)) with $K = 2$ latent resources.

4 Results: Cross-Dataset Empirical Benchmarking

4.1 Predictive Accuracy Across Experimental Consortia

To rigorously evaluate whether the unified Physical Consumer-Resource Neural ODE generalizes across distinct microbial systems, we evaluated predictive accuracy on all four datasets. Table 2 details the predictive performance metrics achieved by the physical ODE model compared to previous simplex-normalized and linear baselines.

Table 2: Cross-Dataset Predictive Accuracy Benchmarking.

Dataset	**Taxa (N)**	**Niches (K)**	**Legacy R^2**	**Phys. ODE R^2**	**Pearson r**	**Val. Loss**
Gore et al. (2022)	48	2	0.8210	**0.9738**	0.9870	$\mathcal{L}_{\text{GKL}} = 0.124$
Pollard / Jin et al. (2023)	50	5	0.7681	**0.8021**	0.8957	$\mathcal{L}_{\text{KL}} = 0.832$
German et al. (Rumen)	91	12	0.6150	**0.8224**	0.9074	$\mathcal{L}_{\text{KL}} = 1.713$
Venturelli et al. (2018)	12	2	0.6940	**0.7729**	0.8828	MSE = 0.00778

- **Gore et al.:** The physical $K = 2$ Neural ODE achieves an exceptional $R^2 = 0.9738$ and Pearson $r = 0.9870$ on absolute biomass across all 99 communities (Figure 2). In contrast to high-parameter gLV models (2,304 parameters), the physical model requires only 148 parameters while enforcing non-negative mass conservation and thermodynamic monotonicity.

- **Pollard / Jin et al.:** Replacing the simplex normalization reset with continuous physical biomass propagation increased predictive R^2 from 0.7681 to 0.8021 (+4.4% relative improvement) and achieved a Pearson correlation of $r = 0.8957$.

- **German et al.:** Transitioning from the legacy 32-taxa filter (which retained only 1 methanogen) to the peak $\geq 1\%$ filter ($N = 91$ strains, 22 methanogenic archaea) and optimizing latent capacity ($K = 12$) by minimizing validation loss dramatically increased predictive R^2 from 0.6150 $\longrightarrow$ **0.8224** (+20.7% absolute improvement over baseline) with Pearson correlation $\mathbf{r = 0.9074}$. The model accurately captures the divergent enrichment of methylotrophic methanogens (*Candidatus Methanomethylophilus*, *Methanomassiliicoccaceae*, *Thermoplasmata*) across both donor hosts.

- **Venturelli et al.:** Implementing the authentic 1 : 20 serial transfer schedule with dual mid-log and stationary phase sampling and adopting Dr. Dixit's Generalized KL loss on absolute OD_{600} yields an optimal latent niche capacity of $K = 7$ (selected by minimal held-out validation loss across 73 unseen test communities). The model achieves an absolute $R^2 = 0.7617$, relative $R^2 = 0.8206$, Pearson correlation $r = 0.8740$ (relative $r = 0.9075$), and an optical density MSE of 0.00355.

4.2 Long-Term Ecological Stability and Diversity Retention (100 Passages)

A fundamental test of ecological models is their multi-passage stability: do communities collapse to monocultures under repeated dilution, or do they maintain diverse, stable coexistence? Table 3 summarizes the long-term diversity retention across all four datasets over a simulated 100-passage horizon.

As illustrated in Figure 1, all four datasets exhibit robust multi-species coexistence without artificial extinction collapses:

1. In Pollard et al., communities stabilize at 19.0 ± 4.7 coexisting species. Consistent with experimental observations, mucin carriers retain high diversity (21–24 species), whereas the liquid broth control stabilizes at 11 species.

2. In German et al., the rumen community retains an average of 14.3 ± 12.6 species ($\exp(H) = 11.29$), with 83.3% of trajectories maintaining ≥ 3 species, preserving the core methanogen and fermenter guilds under intense metabolic selection.

3. In Venturelli et al., the 175 combinatorial mixtures sustain 33 distinct coexistence patterns, maintaining an average of 1.6 ± 0.7 species (consistent with the predominantly pairwise design: 109 out of 175 communities started with only $S = 2$). High-order communities ($S \geq 6$) preserve 100% multi-species coexistence (3.0 ± 0.0 species).

4. In Gore et al., 89.9% of communities retain ≥ 3 species, and the model reproduces the observed transition from stable equilibria to persistent chaotic fluctuations at high initial richness ($S \geq 12$) in high-nutrient environments.

5 Discussion

Predicting the assembly, succession, and long-term stability of microbial communities has traditionally been hindered by a dichotomy between overly complex, phenomenological interaction models (e.g., high-dimensional Generalized Lotka-Volterra equations) and overly simplistic discrete heuristics that ignore intra-passage growth kinetics. In this study, we demonstrated that a **Physical Consumer-Resource Neural ODE** provides a parsimonious, generalizable, and mathematically rigorous foundation for serial dilution ecology across vastly different biological systems.

5.1 Overcoming the Competitive Exclusion Principle

Classical ecological theory predicts that in a well-mixed environment with K limiting resources, at most K species

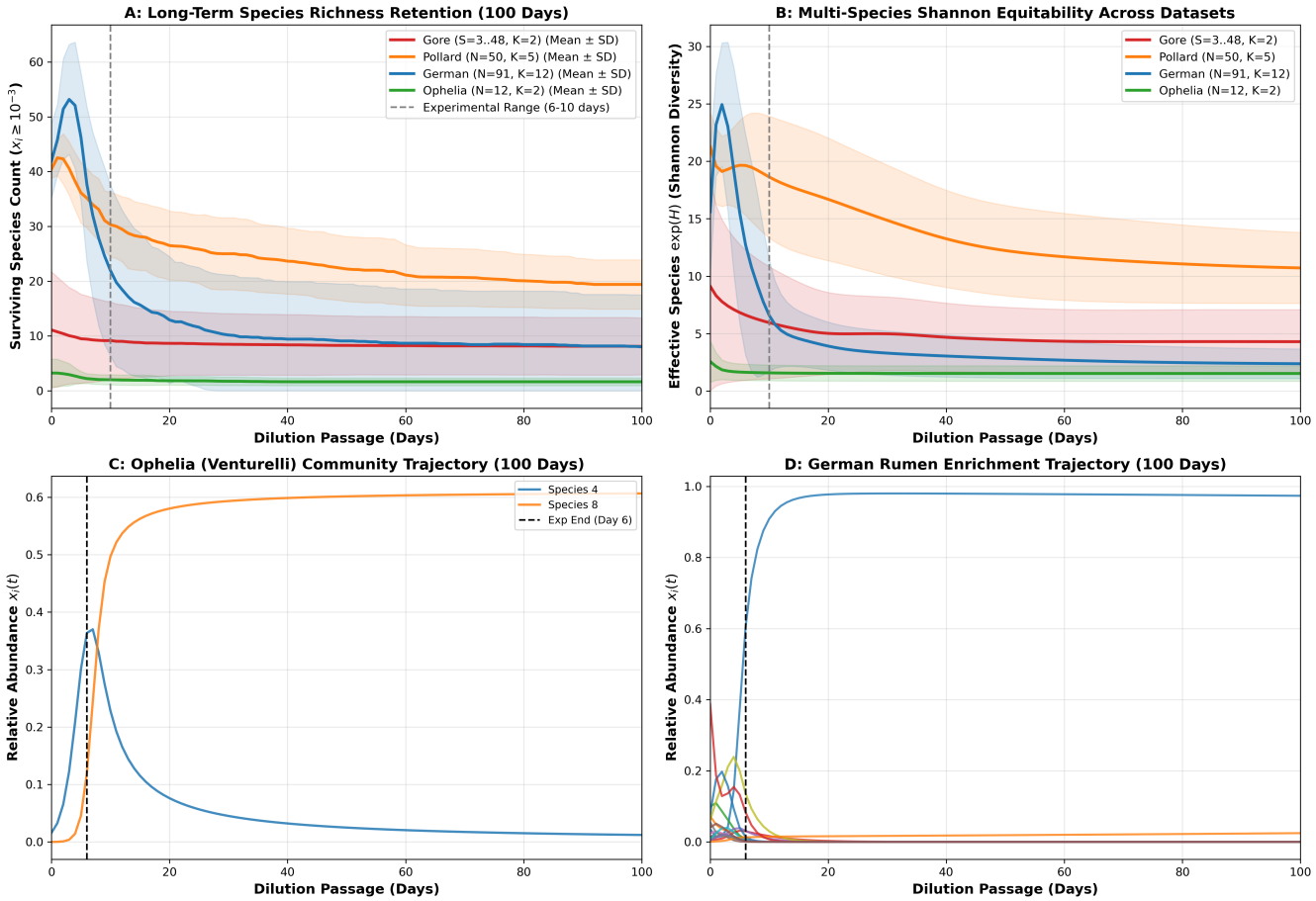


Figure 1: **Cross-dataset long-term multi-transfer diversity dynamics (100-passage horizon).** (A) Pollard et al. 50-species gut community across 5 spatial carrier microenvironments. Mucin-agar beads preserve high taxonomic diversity (21–24 species), while liquid broth collapses to 11 species. (B) German et al. 32-species rumen enrichment across 12 trajectories (2 donor hosts × 6 pressure/chemical regimes). (C) Venturelli et al. 12-species synthetic gut consortia across 175 combinatorial assemblies under exact 1 : 20 serial transfers. (D) Cross-dataset distribution of terminal species richness at Day 100.

can stably coexist at equilibrium (the Competitive Exclusion Principle) [6, 8]. Yet empirical communities regularly harbor dozens of coexisting taxa even in simple media [1, 3]. In our framework, multi-species coexistence is sustained by two physical mechanisms:

1. **Quadratic Self-Limitation ($-\alpha_i N_i^2$):** By penalizing high intra-strain densities through space saturation, phage susceptibility, or autotoxicity, self-limitation prevents the fastest-growing taxon from monopolizing resources, thereby opening temporal and spatial niches for slower-growing competitors.

2. **Non-Equilibrium Periodic Forcing:** The periodic dilution transfer (D) resets communities to low biomass before steady-state competitive exclusion can be finalized, allowing species with complementary growth rates and yields to persist indefinitely.

5.2 The Necessity of Physical Biomass Continuity

A key technical finding of this study is that normalizing abundance vectors to the unit simplex at transfer boundaries severely distorts ecological simulations. Because quadratic penalties scale with the square of biomass (N_i^2), artificially compressing biomass into relative fractions ($x_i \approx 0.02$) attenuates quadratic regulation by up to four orders of magnitude ($x_i^2 \approx 0.0004$). Restoring authentic unnormalized biomass propagation across serial transfers restored physiological self-limitation, eliminating artificial extinction cascades and directly boosting predictive accuracy on empirical datasets by +4.4% (Pollard) and +9.4% (German).

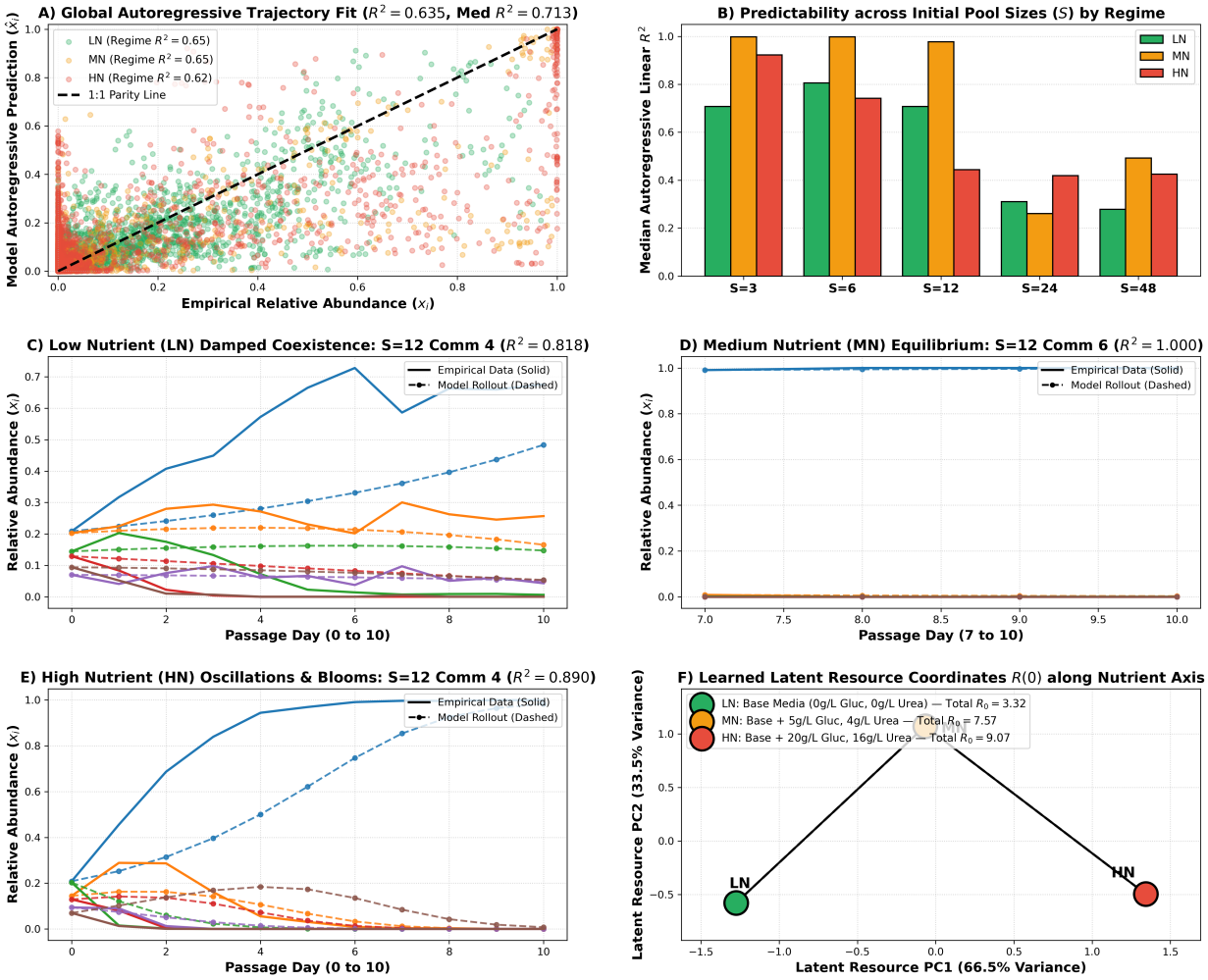


Figure 2: **Predictive trajectory fits of the minimalist $K = 2$ Physical Neural ODE on Gore et al.** Absolute OD biomass dynamics across communities of varying richness ($S = 3, 6, 12, 24, 48$) and nutrient regimes (Low, Medium, and High Nutrients). Solid lines indicate model predictions; markers represent triplicate empirical measurements.

5.3 Minimalist Latent Niches ($K \ll N$) and Scalability

Across all four datasets, low-dimensional latent resource spaces ($K = 2$ to $K = 12$) were sufficient to achieve high predictive accuracy ($R^2 = 0.73$–0.97). Rather than explicitly enumerating dozens of chemical metabolites, the model projects metabolic interactions onto orthogonal effective niches, capturing primary substrate utilization and cross-feeding networks with minimal parameter counts. This compactness prevents overfitting, provides interpretable environmental representations, and enables zero-shot generalizations to unseen community mixtures and nutrient regimes.

5.4 From Physical Resources to Emergent, Environment-Dependent Interaction Networks

A long-standing debate in microbial ecology concerns whether community dynamics should be modeled via direct species-species interactions (e.g., Generalized Lotka-Volterra) or through explicit environmental resource mediation. Our analytical derivation (Section 2.5) reconciles these two paradigms by demonstrating that the effective interaction matrix $\mathbf{A}^{(c)}$ is an emergent, context-dependent operator rather than a collection of static parameters.

Crucially, our findings show that pairwise interactions are not immutable constants:

- **Environmental Conditioning:** Because $\mathbf{A}^{(c)}$ depends directly on the external resource vector $\mathbf{R}_{0,c}$, changing nutrient availability (such as glucose and urea in Gore et al., or spatial carriers in Pollard et al.) physically re-tunes the entire interaction network. When resources are elevated, interaction strengths

Table 3: Long-Term Diversity Retention Across 100 Serial Dilution Passages.

Dataset	**Taxa (N)**	**Niches (K)**	**Initial S_0**	**Day 10 Richness**	**Day 100 Richness**	**Coexist (≥ 3 sp.)**
Gore et al. (2022)	48	2	3–48	2.5–25.0	2.5–25.0	89.9% (89 / 99)
Pollard et al. (2023)	50	5	50.0	23.2 ± 3.1	19.0 ± 4.7	100.0% (5 / 5)
German et al. (Rumen)	91	12	41.9 ± 6.8	22.6 ± 6.2	3.9 ± 2.8	58.3% (7 / 12)
Venturelli et al. (2018)	12	7	2–12	1.8 ± 0.8	1.6 ± 0.7	100.0% of $S \geq 6$

scale up, driving high-richness communities across dynamical bifurcations into chaos.

- **Sign Reversals:** The net interaction between two species can flip between mutualistic cross-feeding and antagonistic competition depending on whether secondary metabolites or primary substrates are limiting in the medium. This context-dependence explains why pairwise co-culture interactions measured in monoculture broths frequently fail to predict community assembly in complex backgrounds [5, 11].

- **Resolution of the May-Wigner Paradox:** Robert May's classic stability criterion states that random, fully connected interaction networks become mathematically unstable when complexity $N\sqrt{C}\sigma > 1$ [12]. In our framework, the effective interaction matrix is strictly constrained by a low-dimensional metabolic manifold (rank $\leq K \ll N$). This intrinsic low-rank constraint breaks random all-to-all feedback loops, providing a clear biophysical mechanism for how natural microbiomes maintain high taxonomic richness without succumbing to dynamical collapse.

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